Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Phillips, Jonathan A Ph.D.		
Project Title:	Genetic Manipulation of Cell Adhesion Proteins to Modify Mechanical Response and Tissue Remodeling		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	01655	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	07/01/2001	**End Date:**	06/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Integrin proteins have been implicated as strong candidates for mediating mechanotransduction in the cell. Integrins are heterodimeric transmembrane proteins, which attach cells to extracellular matrix (ECM), and have been implicated in a variety of signal transduction. It is generally accepted that integrins have the capacity to act as both signal receptors and generators. Integrins can also function synergistically with other receptors to trigger various intercellular signals, resulting in cell proliferation, migration, and contraction. A counter-measure for microgravity related tissue loss would be to heighten the tissue's sensitivity to mechanical input. Since tissue remodeling is a genetic response to mechanical stimulation, changing the signal perceived by the cell should change the remodeling response accordingly. The hypothesis proposed here is that altering the amount of mechanoreceptors (specifically integrins) within tissues can affect signal transduction and therefore cellular response. This hypothesis predicts that increasing the relative amount of integrin protein on a cell can amplify response to environmental mechanical stimuli. Likewise, blocking or down		

	regulating the receptor would quench the signal and therefore the output.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Architecture, density, and composition of load-bearing tissues often adapt to their mechanical environment. Mechanical cues can trigger a myriad of biochemical and genetic responses. Removing ambient mechanical cues, such as in a microgravity environment, is associated with bone loss and compromised mechanical properties. Certain data suggests that load-bearing bones experience more significant changes in microgravity than non-load-bearing bones. The mechanisms by which these changes occur are unknown, however it is intuitive to consider mechanosensitive receptors as major players in these processes. It has been well-established, in spaceflight models and bed rest models, that removing physical cues alters tissue physiology. Elucidating the mechanisms of mechanotransduction and biosynthetic/biophysical response are topics of growing intensity. The applications for genetically altering biomechanical response/sensitivity, extend beyond countermeasures for sustained exposure to microgravity. Being able to change the contractile potential of cells offers a potential therapy for taming aggressive metastatic tumors. Coaxing aggressive tumors into some degree of spontaneous contraction, transiently solidifying the tumor, could make it more "operable" and removable. In contrast, preventing cellular contraction could have therapeutic results in other systems. Inhibiting integrin function through expression of a dominant-negative subunit could have applications in plastic surgery, wound healing, and prevention of scar formation.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)